CREDIT RISK MANAGEMENT ON THE FINANCIAL PERFORMANCE OF AIICO INSURANCE PLC IN NIGERIA OVER THE PERIOD 2010–2023.

BY

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(FMS/INS/20/1032)

DEPARTMENT OF ACTUARIAL SCIENCE

FACULTY OF MANAGEMENT SCIENCE

FEDERAL UNIVERSITY DUTSE, JIGAWA STATE

OCTOBER, 2025

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A PROJECT BEING SUBMITTED TO THE DEPARTMENT OF ACTUARIAL SCIENCES

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IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR THE AWARD OF B.Sc. DEGREE IN INSURANCE

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APPROVAL

This project has been approved for Habibu Bello Lawan whose registration number is FMS/INS/20/1032, by the Department of Actuarial Sciences, Faculty of Management Science, in partial fulfillment of the requirement for the award of Bachelor of Science B.Sc. Degree in Insurance.

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Head of Department Signature & Date

DECLARATION

I Habibu Bello Lawan, hereby declare that this project ‘‘credit risk management on the financial performance of AIICO insurance plc in Nigeria over the period 2010–2023. has been carried out by me under the supervision of Dr. Muhammad Muazu Bala. It has not been presented for award of any degree in any institution. All sources of information are specially acknowledged by means of reference.

Habibu Bello Lawan ……

FMS/INS/20/1032 Date/Sign

DEDICATION

This project work is dedicated to Almighty Allah the Master of the day of Resurrection, to my loving Late Mother Amina Jibril and to my beloved Late Father Bello Abubakar

ACKNOWLEDGEMENT

I remain profoundly grateful to Almighty Allah the creator of the heavens and the earth for His guidance, protection, mercy and grace that kept me all through my project work period. Having mention, the Almighty Allah I cannot but thanks my project supervisor Dr. Muhammad Muazu Bala for always taking his time to go through this work, whom without his support and advice this work will not have been a reality, thank you so much may Almighty Allah bless you and give you more wisdom.

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ABSTRACT

This study investigates the impact of credit risk management on the financial performance of AIICO Insurance Plc in Nigeria over the period 2010–2023. Three key credit risk indicators liquidity ratio, capital adequacy ratio, and short leverage debt ratio were examined in relation to financial performance, measured by return on assets (ROA). Using a longitudinal case study design, data were sourced exclusively from AIICO's audited financial statements and analyzed with correlation and regression techniques. The findings reveal that liquidity ratio and capital adequacy ratio had negative and weak positive but statistically insignificant effects on financial performance, respectively. In contrast, short leverage debt ratio exhibited a strong and significant positive relationship with performance, emerging as the most influential determinant. The overall regression model demonstrated high explanatory power ($R^2 = 0.894$), indicating that the selected indicators strongly predict firm performance. The study concludes that while liquidity and capital adequacy serve regulatory and solvency functions, short-term leverage plays a central role in driving profitability within AIICO. The research provides valuable insights for insurance firms, regulators, investors, and policymakers, emphasizing the need for strategic debt utilization, efficient liquidity deployment, and balanced capital requirements to ensure sustainable growth and resilience in the Nigerian insurance sector.

Keywords: Credit Risk Management, Financial Performance, AIICO Insurance Plc, Leverage Ratio, Capital Adequacy

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CHAPTER ONE

INTRODUCTION

1.1 Background to the Study:

Credit risk management is unequivocally a core component of financial stability, particularly within the dynamic and capital-intensive insurance sector. This industry, globally and in Nigeria, inherently relies on effective capital structures and prudent risk practices to fulfil its primary function of indemnifying policyholders against unforeseen losses (Cummins & Phillips, 2011; NAICOM, 2024a). In Nigeria, the insurance industry plays a pivotal role in economic development, serving as a significant financial intermediary by mobilizing long-term funds for investment and providing essential risk coverage for individuals, businesses, and government entities (CBN, 2023). Its contribution extends to fostering economic resilience and facilitating enterprise by mitigating uncertainties that could otherwise hinder investment and innovation (Echekoba et al., 2017).

despite its crucial developmental role, the Nigerian insurance sector has historically grappled with persistent challenges that often impede its optimal performance and contribution to the national economy. These issues include a perceived undercapitalization relative to more developed markets, the prevalence of suboptimal or evolving risk m anagement frameworks, and recurrent liquidity constraints that can compromise claims payment and operational efficiency (PwC, 2022; Fapetu & Ogege, 2019). These factors cumulatively impact the operational viability, profitability, and overall public perception of insurance companies, often leading to inhibited market penetration and growth (Adeleke et al., 2015). The solvency and stability of these firms are paramount, as their failure can have ripple effects across the financial system, eroding public confidence and potentially creating systemic risks (NAICOM, 2024b).

Credit risk, fundamentally, represents the potential for financial loss arising from a counterparty's failure to meet its contractual obligations, be it a borrower, an investment issuer, or a business partner (Hull, 2018). For insurance companies, credit risk primarily manifests in several key areas. Foremost is the default risk associated with their extensive investment portfolios, which are essential for generating returns to meet future claims and

often comprise government securities, corporate bonds, equities, and real estate (Babalola, 2012; Olayinka & Onuoha, 2019). Secondly, there is significant counterparty risk with reinsurance companies, where the reinsurer might fail to honor its share of claims. Lastly, the risk of non-payment of premiums by policyholders also constitutes a form of credit risk (Adegbite, 2020). Effective credit risk management is therefore non-negotiable for insurance firms to remain solvent and profitable. This involves a suite of proactive measures such as maintaining adequate capital buffers to absorb unexpected losses, ensuring sufficient liquidity to meet immediate claims and operational needs, and implementing stringent credit underwriting standards for both investments and premium receivables (Gitman & Zutter, 2018).

The period from 2010 to 2023 has been particularly transformative for Nigeria's insurance industry. This era has witnessed significant regulatory adjustments, notably the continued implementation of the National Insurance Commission (NAICOM) Act and various prudential guidelines aimed at strengthening financial health and enhancing risk controls across the sector (NAICOM, 2010-2023 Annual Reports). Key reforms included increases in minimum capital requirements and revised solvency regulations, which pushed insurers to review their capital structures and risk management practices (NAICOM, 2021 Guidelines). Furthermore, the Nigerian economy experienced considerable macroeconomic volatilities, including periods of high inflation, currency devaluation, fluctuating oil prices, and the global impact of the COVID-19 pandemic, which tested the resilience of all financial institutions (NBS, 2024; KPMG, 2022). These external pressures directly influenced the investment landscape, the credit quality of counterparties, and the overall operational environment for insurers. Consequently, the robustness and adaptability of credit risk management practices became even more critical in navigating these turbulent times.

This study specifically focuses on AIICO Insurance Plc, a prominent and long-standing player in the Nigerian insurance market, listed on the Nigerian Exchange Group (NGX). AIICO has a diversified business model, offering both life and non-life insurance products, and holds a significant share of the market, making it an ideal case for in-depth analysis (AIICO Annual Reports, 2010-2023). By examining how critical credit risk indicators namely, liquidity ratio, capital adequacy ratio, and short leverage debt ratio influence AIICO's

financial performance over this thirteen-year period (2010-2023), this research aims to provide an in-depth, firm-specific understanding. Such a focused case study can reveal unique insights into the operationalization and efficacy of credit risk management strategies within a leading Nigerian insurer, contributing valuable empirical evidence to the discourse on financial stability in emerging markets.

1.2 Statement of the Problem:

Despite the critical role Nigerian insurance companies play in financial intermediation, risk mitigation, and overall economic stability, the sector has consistently faced challenges leading to a perception of underperformance or slower growth compared to its potential (Fapetu & Ogege, 2019; PwC, 2022). This underperformance is frequently, and strongly, linked to deficiencies in credit risk management, which can expose firms to significant vulnerabilities and ultimately compromise their ability to meet obligations (Adegbite, 2020). Persistent issues such as liquidity crises, inadequate capital buffers to absorb unexpected losses from investment defaults, and an over-reliance on often expensive and volatile short-term debt sources raise serious concerns about their operational viability and long-term sustainability (Adamu & Ibrahim, 2015). These vulnerabilities can lead to compromised claims settlement, erosion of policyholder confidence, and ultimately, systemic risk within the broader financial sector (NAICOM, 2024b).

While existing academic literature has explored the various aspects of risk management in Nigerian financial institutions, a notable empirical and contextual gap persists. Many prior studies tend to focus predominantly on the banking sector, whose credit risk exposures and management frameworks, though sharing common principles, differ significantly from those of insurance companies due to their distinct business models and regulatory environments (e.g., Olayinka & Onuoha, 2019; Babalola, 2012). Furthermore, studies specifically on Nigerian insurance firms often examine individual risk indicators in isolation, rather than investigating their combined and interactive effects within a comprehensive credit risk management framework (e.g., Adegbite, 2020, focusing solely on liquidity). Crucially, there is a significant void in the analysis of how the interplay of specific credit risk metrics liquidity ratio, capital adequacy ratio, and short leverage debt ratio

impacts the financial performance of a single, prominent Nigerian insurance firm over an extended period that encompasses diverse economic cycles and significant regulatory adjustments, such as the 2010-2023 timeframe.

This prolonged period witnessed major macroeconomic shifts (e.g., recessions, oil price volatility) and regulatory reforms (e.g., capital base increases, IFRS 17 adoption in 2023) that likely tested the robustness of insurers' credit risk management capabilities and capital resilience (KPMG, 2022; NAICOM, 2023 Guidelines). The absence of an in-depth, longitudinal case study focusing on a major player like AIICO Insurance Plc leaves a critical gap in understanding the nuanced, firm-specific relationship between proactive risk management and sustainable performance in this unique and evolving market. This research therefore aims to fill this specific empirical and contextual gap by providing a comprehensive, firm-specific analysis of AIICO Insurance Plc, thereby offering insights into the challenges and opportunities in credit risk management within a leading Nigerian insurance company.

1.3 Objectives of the Study:

The main objective of this study is to assess the impact of credit risk management on the financial performance of AIICO Insurance Plc in Nigeria from 2010 to 2023.

The specific objectives are:

1. To examine the effect of liquidity ratio on the financial performance of AIICO Insurance Plc.
2. To assess the relationship between capital adequacy ratio and the financial performance of AIICO Insurance Plc.
3. To evaluate the impact of short leverage debt ratio on the financial performance of AIICO Insurance Plc.

1.4 Research Questions:

1. What is the relationship between liquidity ratio and the financial performance of AIICO Insurance Plc in Nigeria from 2010 to 2023?
2. How does capital adequacy ratio influence the financial performance of AIICO Insurance Plc from 2010 to 2023?

3. What effect does short leverage debt ratio have on the financial performance of AIICO Insurance Plc from 2010 to 2023?

1.5 Research Hypotheses:

H_{11}: Liquidity ratio has significant effect on the financial performance of AIICO Insurance Plc.

H_{12}: Capital adequacy ratio does has significantly affect the financial performance of AIICO Insurance Plc.

H_{13}: Short leverage debt ratio has significant impact on the financial performance of AIICO Insurance Plc.

1.6 Significance of the Study:

This study holds significant implications and provides valuable insights for a diverse range of stakeholders within the Nigerian financial ecosystem and beyond:

i. For AIICO Insurance Plc and Other Insurance Companies: The empirical findings will offer crucial evidence on how specific credit risk management practices, as proxied by the selected financial ratios, directly influence their financial health, profitability, and long-term sustainability. This will empower AIICO's management with data-driven insights to refine their credit risk policies, optimize investment strategies, improve capital allocation decisions, and enhance overall risk mitigation frameworks, leading to improved operational efficiency and resilience. The detailed case study of AIICO can also serve as a learning benchmark for other insurance companies in Nigeria struggling with similar credit risk challenges.

ii. For Regulators (e.g., National Insurance Commission - NAICOM): The study's outcomes can serve as a vital input for policy formulation and refinement. By identifying the critical links between specific credit risk metrics (liquidity, capital adequacy, short-term leverage) and firm performance, NAICOM can develop more targeted, effective prudential guidelines, solvency regulations, and supervisory frameworks that truly reflect the operational realities and risk profiles of Nigerian insurers. This will contribute to fostering greater financial stability, protecting policyholders' interests, and ensuring the sound and sustainable growth of the entire insurance sector in Nigeria.

iii. For Investors and Stakeholders: The research will provide a clearer, empirically-backed understanding of the credit risk profile and financial performance drivers within a leading Nigerian insurance company like

AIICO. This granular insight will facilitate more informed investment decisions, enable a more accurate assessment of risk-adjusted returns, and enhance due diligence processes for potential and existing shareholders, bondholders, and other financial stakeholders considering the Nigerian insurance market.

iv. For Researchers and Academics: This study makes a distinct contribution to the existing body of financial risk management literature, particularly within the context of emerging markets. By addressing identified gaps concerning the long-term impact of specific credit risk metrics on the performance of a case study insurance firm, it offers novel empirical evidence that can be used for comparative studies or as a foundation for further research on financial risk management, capital adequacy, liquidity, and leverage within the insurance industry, particularly in developing economies. It adds to the understanding of how regulatory changes and macroeconomic shocks affect firm-level performance.

1.7 Scope of the Study:

This study is specifically limited to a detailed case study of AIICO Insurance Plc, which is recognized as one of the prominent and publicly listed insurance companies operating in Nigeria. The research focuses exclusively on the financial performance and credit risk management indicators of AIICO Insurance Plc over a defined thirteen-year period, spanning from January 1, 2010, to December 31, 2023. This timeframe is selected to capture a substantial period of operational history, encompassing various economic cycles, significant regulatory reforms (e.g., NAICOM recapitalization directives, IFRS 17 adoption), and industry-specific developments within Nigeria.

The primary emphasis of this research is on assessing the impact of three key credit risk indicators namely, the liquidity ratio (Current Assets/Current Liabilities), capital adequacy ratio (Total Equity/Total Assets), and short leverage debt ratio (Short-term Borrowings/Total Assets) on AIICO's financial performance, as measured by Return on Assets (ROA). Data collection will primarily rely on the company's audited annual financial statements and reports made publicly available during this period. While acknowledging the multifaceted nature of risk in insurance, this study will not delve into micro-level operational risks, specific underwriting risks, or detailed actuarial valuations, except where these indirectly manifest or are explicitly reported as credit risk

exposures relevant to the defined variables. The findings and conclusions drawn will therefore be specific to AIICO Insurance Plc during the specified period and, while offering deep insights, may require careful consideration for generalization to the broader Nigerian insurance industry.

1.8 Limitations of the Study

1. Firstly: the research is restricted to a single case study of AIICO Insurance Plc. While this allows for an in-depth and firm-specific analysis, the findings may not be fully generalizable to other insurance companies in Nigeria or across different markets, as each firm operates under unique organizational structures, strategies, and risk exposures.

2. **Secondly:** the study focuses on only three credit risk indicators liquidity ratio, capital adequacy ratio, and short leverage debt ratio while other potential determinants of financial performance such as underwriting risks, market risks, operational risks, and macroeconomic shocks are not included in the model.

1.9 Organization of the Study

1. **Chapter One:** introduces the study by providing the background, problem statement, objectives, research questions, hypotheses, significance, scope, and the identified limitations.
2. **Chapter Two:** presents the literature review, which covers the conceptual, theoretical, and empirical frameworks, and highlights the gap this study seeks to fill.
3. **Chapter Three:** focuses on the research methodology, detailing the research design, population, sample size, data sources, data collection techniques, and methods of data analysis and model specification employed.
4. **Chapter Four:** presents the results and analysis of the study. It includes the presentation of data, interpretation of findings, and discussions in relation to the research objectives and questions.
5. **Chapter Five:** provides the summary of major findings, conclusions drawn from the study, and recommendations for policy, practice, and further research.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction:

The literature review forms the theoretical and empirical foundation for this study, establishing a clear understanding of credit risk management, insurance company performance, and the relationships between them. This chapter is structured into conceptual, theoretical, and empirical frameworks, culminating in the identification of the specific gap this research aims to address.

2.2 Conceptual Framework:

This section defines and elaborates on the key concepts central to this study, providing a shared understanding of the terminology and their relevance within the insurance context.

a. **Credit Risk Management:** Credit risk management encompasses the systematic policies, procedures, and practices employed by an organization to identify, measure, monitor, and control potential losses arising from a counterparty's failure to meet its financial obligations (Hull, 2018). For insurance companies, credit risk primarily stems from:

   i. Investment Portfolio: Insurers hold substantial investment portfolios to generate returns and ensure liquidity for claims. Default on bonds, non-performing loans, or devaluation of equity investments due to issuer financial distress are core credit risks (Saunders & Cornett, 2019).

   ii. Reinsurance Counterparty Risk: This is the risk that a reinsurer fails to pay its share of ceded claims, leaving the primary insurer with unexpected liabilities (Rejda & McNamara, 2020).

   iii. Premium Receivables: The risk of policyholders or brokers failing to pay premiums due can directly impact an insurer's cash flow and profitability (Gitman & Zutter, 2018).

   iv. Effective credit risk management in insurance involves robust credit assessments, diversification strategies, setting exposure limits, collateral management, and continuous monitoring of counterparties' creditworthiness (Khan & Ahmad, 2012).

b. **Liquidity Ratio:** A liquidity ratio is a financial metric that indicates a company's ability to meet its short-term financial obligations using its most liquid assets (Brigham & Houston, 2019). For insurance companies, maintaining adequate liquidity is critical for meeting immediate claims payments, operational expenses, and unexpected cash outflows.

   i. The Current Ratio (Current Assets / Current Liabilities) is a commonly used liquidity indicator. A higher current ratio generally suggests a stronger ability to cover short-term liabilities without resorting to forced asset sales or expensive short-term borrowing (Kalu & Mgbemena, 2019). However, excessively high liquidity might indicate inefficient asset utilization.

   ii. In the context of insurers, careful consideration is given to the composition of current assets (e.g., highly marketable securities vs. less liquid assets) and the nature of current liabilities (e.g., short-term claims payable vs. operating payables).

c. **Capital Adequacy Ratio (CAR):** This ratio measures a firm's financial strength and its capacity to absorb unexpected losses, particularly those arising from various risks including underwriting, market, operational, and crucially, credit risks (NAICOM, 2024c). For insurance companies, CAR reflects the sufficiency of their capital base relative to their risk exposures.

   i. While banks typically use a risk-weighted assets (RWA) approach (Basel III), insurance regulators like NAICOM often prescribe specific solvency margins or capital requirements based on premium income or total assets (NAICOM Act 2003).

   ii. In the absence of publicly available "risk-weighted assets" for an insurer, the ratio of Total Equity to Total Assets serves as a widely accepted proxy for capital adequacy. This ratio indicates the proportion of a company's assets financed by equity, reflecting its long-term financial stability and buffer against unforeseen losses (Brealey, Myers, & Allen, 2020). A higher capital adequacy implies greater resilience against credit defaults in its investment portfolio and other financial shocks.

d. **Short Leverage Debt Ratio:** This ratio reflects the proportion of a company's assets financed by short-term liabilities (excluding operational liabilities like claims payable, focusing on debt-like instruments). It

indicates an insurer's reliance on short-term funding sources which can be volatile and more susceptible to interest rate fluctuations (Gitman & Zutter, 2018).

i. Specifically, the ratio of Short-term Borrowings to Total Assets is used here. A high short leverage debt ratio may suggest increased financial risk, especially if these short-term funds are used to finance long-term, illiquid assets, potentially leading to maturity mismatches and heightened liquidity concerns (Modigliani & Miller, 1958). For an insurer, this could imply greater pressure to generate quick returns or face refinancing risks.

e. **Financial Performance:** For insurance companies, financial performance is a multi-faceted concept, typically measured using profitability ratios that indicate the efficiency and effectiveness with which the company generates returns from its operations and investments (Kalu & Mgbemena, 2019).

i. Return on Assets (ROA): Calculated as Net Income divided by Total Assets (Net Income / Total Assets). ROA is a key indicator of how efficiently a company is utilizing its assets to generate profit (Brigham & Houston, 2019). For insurers, a higher ROA implies effective management of both underwriting and investment portfolios to maximize overall returns relative to the size of their asset base. It is a robust measure that captures the holistic performance derived from both operational activities and investment income, which is crucial for an insurance company managing large asset pools.

2.2 Theoretical Framework:

The theoretical framework provides the conceptual lens through which the relationship between credit risk management and insurance company performance will be analysed. This study draws primarily from Modern Portfolio Theory, Risk Management Theory, and aspects of Capital Structure Theory.

2.2.1 Modern Portfolio Theory (MPT) by Markowitz (1952)

i. Core Tenets: Developed by Harry Markowitz, MPT asserts that investors can reduce risk for a given level of expected return (or maximize return for a given level of risk) by combining different assets into a diversified portfolio. The theory emphasizes that the risk of a portfolio is not merely the sum of the risks of

its individual assets but also depends on how these assets covary (Markowitz, 1952). Diversification reduces unsystematic (specific) risk.

ii. Relevance to Study: In the context of credit risk management for insurance companies, MPT highlights the critical importance of diversifying investment exposures. Insurers hold substantial asset portfolios, and credit risk arises from the potential default of these assets (e.g., bonds, money market instruments). By diversifying across different asset classes, industries, geographic regions, and counterparties with varying credit profiles, insurers can mitigate the impact of specific credit defaults on their overall portfolio performance (Khan & Ahmad, 2012). MPT provides the theoretical underpinning for understanding how effective portfolio diversification, a key aspect of credit risk management, can lead to more stable investment returns and, consequently, enhance an insurer's overall financial performance and solvency by minimizing unexpected credit losses.

2.2.2 Risk Management Theory:

i. Core Tenets: Risk management theory encompasses a broad set of principles and frameworks for identifying, assessing, controlling, monitoring, and mitigating various forms of risk that an organization faces. Its fundamental premise is that effective risk management enables an organization to achieve its objectives by systematically managing uncertainties (Smith & Merritt, 2002). This involves a continuous process of proactive identification of potential threats, quantitative and qualitative assessment of their likelihood and impact, development of strategies to reduce or transfer risk, and ongoing monitoring of risk exposures and control effectiveness.

ii. Relevance to Study: This theory is central to understanding how insurance firms identify, measure, mitigate, and monitor their credit risk exposures to optimize financial performance. It provides the foundation for the various credit risk management practices implemented by AIICO Insurance Plc, such as establishing credit policies, assessing counterparty creditworthiness, setting exposure limits, and making provisions for doubtful debts (Adegbite, 2020). The theory posits that robust risk management practices lead to reduced volatility in earnings, improved solvency, and enhanced profitability by minimizing unexpected losses. The

study's focus on liquidity, capital adequacy, and short leverage debt ratios directly relates to managing the financial risks, including credit risks, that an insurer faces, aligning with the objectives of sound risk management theory.

2.2.3 Capital Structure Theory (Modigliani and Miller, 1958; including subsequent extensions)

i. Core Tenets: Initially developed by Modigliani and Miller (M&M), this theory explores the relationship between a firm's capital structure (mix of debt and equity) and its value and cost of capital. While their initial propositions suggested that capital structure is irrelevant under perfect market conditions, subsequent extensions (e.g., by Miller himself and others, considering taxes, financial distress costs, and agency costs) demonstrated that an optimal capital structure can exist (Myers, 1984). High leverage can increase expected returns to equity but also increases financial risk, including the probability of default.

ii. Relevance to Study: In the context of insurance companies, Capital Structure Theory is highly relevant when considering their capital adequacy and leverage, particularly the reliance on short-term debt. The theory provides a lens to analyze how the level and composition of an insurer's funding (equity vs. debt, and the maturity of debt) can influence its cost of capital, its solvency, its ability to absorb credit losses from investments, and ultimately, its financial performance (Adamu & Ibrahim, 2015). A high short leverage debt ratio, for instance, might indicate an aggressive financing strategy that, while potentially boosting returns in good times, could expose the insurer to significant refinancing risk and financial distress costs if credit markets tighten or if asset quality deteriorates.

2.3 Empirical Review:

This section critically reviews relevant empirical studies that have investigated the relationship between credit risk management and financial performance, both globally and specifically within the Nigerian context. The review highlights methodologies, key findings, and identified limitations, leading to the articulation of the current study's unique contribution.

2.3.1 Studies on Capital Structure, Capital Adequacy, and Firm Performance:

i. Ujunwa & Modebe (2011), in their study "Capital Structure and Firm Performance: Evidence from Nigerian Quoted Companies," published in the Journal of Finance and Economics, examined the impact of capital structure on the performance of a sample of Nigerian quoted firms. Their findings generally suggested that the mix of debt and equity influences profitability, though the specific relationship varied across industries. While not exclusive to insurance, their work highlighted the importance of leverage in the Nigerian corporate landscape.

ii. Adamu & Ibrahim (2015), in "Capital Structure and Financial Performance of Insurance Companies in Nigeria," published in the Journal of Economics and Sustainable Development, directly investigated the relationship for Nigerian insurers. They found that certain capital structure components, including debt, had significant effects on financial performance metrics like profitability. This study provides direct empirical support for the relevance of capital adequacy and leverage in the Nigerian insurance sector.

iii. Limitations of existing capital structure studies: While these studies confirm the link between capital structure and performance, they often do not isolate "credit risk management" as the primary driver, nor do they typically delve into specific components like the "short leverage debt ratio" as a distinct credit risk indicator. Furthermore, a long-term, single-firm case study of this nature remains limited.

2.3.2 Studies on Liquidity Management and Performance:

i. Adegbite (2020), in "Impact of Liquidity Management on the Financial Performance of Insurance Companies in Nigeria," found a significant positive relationship between liquidity and the profitability of Nigerian insurance companies. Using proxies like the current ratio, the study concluded that effective liquidity management is crucial for operational efficiency and meeting obligations, which in turn enhances performance.

ii. Similarly, Ogundipe, Olayiwola, & Adeyemi (2014), in their work "Liquidity Risk Management and Financial Performance of Insurance Companies in Nigeria," also reported a positive association between liquidity risk management practices and the financial performance of insurers. They highlighted that

maintaining optimal liquidity buffers enables insurers to withstand unforeseen shocks and capitalize on investment opportunities.

iii. Limitations of existing liquidity studies: While these studies strongly establish the importance of liquidity, they often treat it as a singular aspect of financial management rather than integrating it explicitly within the broader framework of credit risk management. They also tend to employ panel data across multiple firms, which might obscure firm-specific nuances, and typically cover shorter timeframes than the 2010-2023 period.

2.3.3 Studies on Credit Risk and Profitability (General Financial Institutions)

i. Olayinka & Onuoha (2019), in their study on "Credit Risk Management and Profitability of Deposit Money Banks in Nigeria," used measures like non-performing loans and loan loss provisions as proxies for credit risk. They found that effective credit risk management significantly and positively influenced the profitability of Nigerian deposit money banks. Their methodology for quantifying credit risk can provide insights for adapting to the insurance context, even though the primary source of credit risk differs.

ii. Limitations of credit risk studies (general financial institutions): These studies provide a foundational understanding of credit risk's impact on financial performance. However, their primary focus on banks means their direct applicability to insurers is limited due to differences in credit exposures (e.g., investment defaults versus loan defaults) and regulatory frameworks. They rarely delve into the specific ratios like capital adequacy relative to credit risk or the nuanced role of short-term leverage within an insurer’s unique balance sheet structure over an extended period.

2.4 Gap in the Literature:

The comprehensive review of existing empirical literature reveals several key limitations and a significant gap that this study aims to address:

1. Sectoral Focus Discrepancy: A substantial portion of research on financial risk management and performance in Nigeria predominantly focuses on the banking sector. While valuable, these findings are

not entirel y transferable to the insurance industry due to fundamental differences in their business models, primary sources of credit risk (investment portfolio vs. loan portfolio), and distinct regulatory frameworks.

2. Lack of Integrated Credit Risk Metrics: Many prior studies, even those on insurance firms, tend to examine individual financial ratios (e.g., liquidity or capital adequacy) in isolation, rather than explicitly integrating them as components of a holistic "credit risk management" framework. There is a dearth of research that simultaneously investigates the combined effects of liquidity ratio, capital adequacy ratio, and short leverage debt ratio as crucial indicators of an insurer's credit risk exposure and its subsequent impact on performance.
3. Limited Long-Term, Firm-Specific Analysis: Most existing studies utilize cross-sectional data or shorter timeframes, often analysing panel data across multiple firms. While useful for generalized conclusions, this approach often overlooks the unique, firm-specific dynamics and nuances of risk management evolution within a single prominent company over an extended period. The 2010-2023 timeframe for AIICO Insurance Plc, encompassing various economic cycles and significant regulatory shifts in Nigeria, provides a unique opportunity for an in-depth longitudinal analysis that is currently absent in the literature.
4. Absence of a Dedicated Case Study: There is a notable absence of a detailed, empirical case study focusing specifically on how a leading Nigerian insurer like AIICO Insurance Plc manages its credit risk exposures (through the identified ratios) and how these practices translate into its long-term financial performance. Such a study can provide granular insights that broad-brush industry analyses cannot capture.

This study, therefore, aims to fill these identified gaps by providing a focused, long-term (2010-2023) panel data analysis of AIICO Insurance Plc. By examining the intertwined effects of liquidity ratio, capital adequacy ratio, and short leverage debt ratio on its financial performance (ROA), this research will contribute unique, firm-specific, and contextually rich empirical evidence to the existing body of knowledge on financial risk management within the Nigerian insurance industry

CHAPTER THREE

RESEARCH METHODOLOGY

3.0 introduction:

This chapter outlines the systematic approach that will be employed to investigate the impact of credit risk management on the financial performance of AIICO Insurance Plc. It details the research design, clarifies the population and sample, identifies the data sources, defines and measures the variables, specifies the analytical model, and describes the estimation techniques to be used. Measures for ensuring validity and reliability, alongside ethical considerations, are also addressed.

3.1 Research Design:

This study adopts a quantitative research design, specifically a longitudinal case study approach utilizing Correlation and Regression analysis. This design is chosen for its suitability in evaluating the dynamic relationship between credit risk indicators and financial performance within a single, prominent entity (AIICO Insurance Plc) over an extended period (2010-2023).

a) Quantitative Research: The primary aim of this research is to establish numerical relationships and test hypotheses using measurable variables derived from AIICO's financial statements. This necessitates a quantitative approach, which allows for objective measurement, statistical analysis to determine the strength and significance of relationships, and the testing of the formulated hypotheses (Creswell & Creswell, 2018).

b) Longitudinal Case Study: Focusing exclusively on AIICO Insurance Plc across a thirteen-year period (2010-2023) constitutes a longitudinal case study. This approach enables an in-depth, firm-specific analysis of the company's financial performance and its credit risk management practices over time (Yin, 2018). It allows for the observation of trends, the impact of specific internal strategies, and the responses to external macroeconomic shifts and regulatory changes. This temporal dimension is critical for understanding the dynamic interplay between risk management practices and financial outcomes within a real-world organizational context.

3.2 Population and Sample Size:

The population for this study comprises only AIICO insurance plc operating in Nigeria from 2010 to 2023. In alignment with the in-depth nature of a case study design, the sample for this research is exclusively AIICO Insurance Plc. AIICO Insurance Plc has been purposefully selected as the focus of this study due to several strategic considerations:

a) Market Prominence and Longevity: AIICO Insurance Plc is a well-established and leading player in the Nigerian insurance industry, with a substantial market presence across both life and non-life insurance segments. Its long history of operations (since 1963) ensures a rich dataset for a longitudinal analysis and provides insights relevant to a significant entity within the sector.

b) Public Listing and Data Accessibility: As a publicly listed company on the Nigerian Exchange Group (NGX), AIICO Insurance Plc is legally mandated to publish comprehensive, audited annual financial reports. This ensures consistent, verifiable, and publicly accessible data for the entire study period (2010-2023), which is fundamental for the reliability and validity of the quantitative analysis.

c) Diversified Operations: AIICO's diversified portfolio across various insurance products and services offers a holistic view of an insurer's financial operations, providing a robust context for analysing the multifaceted relationship between credit risk and performance within a single, complex organization. The selection of a single, well-documented case allows for a deep, intensive investigation of the complex phenomenon of credit risk management within its specific organizational and market context (Yin, 2018).

3.3 Sources of Data:

This study will rely exclusively on secondary data directly sourced from AIICO Insurance Plc's publicly available financial records. No other company's financial data or aggregated industry data will be used for the primary variables. The primary source of data will be the audited annual financial reports of AIICO Insurance Plc for the financial years 2010 to 2023. These reports are official, regulated financial disclosures and are considered highly reliable for quantitative analysis. They are publicly accessible via AIICO's official investor

relations website (e.g., www.aiicoplc.com/about-us/investor-relations) and through the archives of the Nigerian Exchange Group (NGX) website (www.ngxgroup.com).

Specifically, data will be meticulously extracted from:

a. Statement of Financial Position (Balance Sheet): This will be the primary source for balance sheet items. Data points to be extracted include:
    i. Total Assets: The aggregate value of all assets owned by the company.
    ii. Current Assets: Assets expected to be converted into cash or used within one year (e.g., cash and bank balances, short-term investments, trade receivables, reinsurance assets).
    iii. Current Liabilities: Obligations expected to be settled within one year (e.g., insurance contract liabilities, trade payables, other financial liabilities, current tax liabilities).
    iv. Total Equity: The residual interest in the assets of the company after deducting all its liabilities (share capital, retained earnings, reserves).
    v. Short-term Borrowings: Financial liabilities classified as current, representing loans or debt obligations due for repayment within one year. This will be specifically identified within the current liabilities section or relevant notes.
b. Statement of Profit or Loss and Other Comprehensive Income (Income Statement): This will be the primary source for profitability figures. The key data point to be extracted is:
    i. Net Income (Profit After Tax): The final profit figure reported by the company after all expenses, including taxes, have been deducted from revenue.
c. Notes to the Financial Statements: These accompanying notes are critical for providing detailed breakdowns, specific definitions of accounting policies, and explanations of significant balances. They will be consulted to ensure accurate identification and extraction of data for variable calculation, and to understand any changes in accounting standards or classifications over the 13-year period (e.g., the potential impact of IFRS 17 adoption on certain revenue or liability components from 2023 onwards, ensuring comparability for the core variables used).

While external industry reports or regulatory guidelines (e.g., from NAICOM) may be consulted for contextual understanding of the Nigerian insurance operating environment or to identify general industry trends, they will not be used as sources for the raw financial data of AIICO Insurance Plc itself. All quantitative data points for the variables defined below will strictly originate from AIICO's own audited annual reports.

3.4 Variables and Measurement:

The study will involve one dependent variable and three independent variables, all measured using financial ratios derived from the audited annual financial statements of AIICO Insurance Plc. The consistent application of these definitions across the 13-year period is crucial for data reliability.

3.5 Validity and Reliability of the Study

The reliability of this study is assured by the use of audited, publicly available financial statements. As a company listed on the AIICO Insurance plc website and Nigerian Exchange Group (NGX), is subject to rigorous reporting standards imposed by regulatory bodies, this regulatory oversight ensures the accuracy and trustworthiness of the financial data.

1. Direct Source Data: The data is extracted directly from the company's own official reports, eliminating the risk of third-party interpretation or data manipulation.
2. Consistency: Standard definitions for financial terms and consistent data structures are used across all fourteen years, ensuring a uniform basis for analysis.
3. Standardized Techniques: The application of standard and widely accepted financial analysis techniques (multiple linear regression, correlation analysis,) ensures that the findings are methodologically sound and verifiable.

3.6 Ethical Considerations

1. Public Data: All data used is in the public domain, so there is no need to seek permission or access confidential company information.
2. Academic Purpose: The study is conducted strictly for academic purposes and is not intended to influence investor decisions, manipulate stock prices, or harm the company’s reputation.
3. Transparency: Proper citations and a bibliography will be provided for all sources of data and information to ensure transparency and prevent plagiarism. The methodology is laid out in detail so that others can understand and potentially replicate the study.

3.7 Method of Analysis:

This section details the statistical techniques and analytical framework that will be employed to evaluate the relationship between credit risk management proxies and the financial performance of AIICO Insurance Plc over the period 2010 to 2023.

3.7.1 Regression analysis

3.7.2 Correlation Analysis

3.8 Model Specification:

To access the effect of credit risk indicators on financial performance, the following regression model is specified'

$$FPt = \beta o + \beta 1\ LIQRt + \beta 2\ CARt + \beta 3\ STLDRt + \varepsilon t$$

Where:

FPt: Financial performance (e.g., ROA or Net Income)

LIQRt: Liquidity Ratio

CARt: Capital Adequacy Ratio

STLDRt: Short-Term Leverage Debt Ratio

βo: Intercept

$\beta 1, \beta 2, \beta 3$: Coefficients

εt: Error term

CHAPTER FOUR

Results and Findings

4.1 Introduction

This chapter presents the results of the statistical analysis carried out to examine the relationship between liquidity ratio, capital adequacy ratio, short leverage debt ratio, and performance. Both correlation and regression analyses were performed using SPSS, and the findings are presented below.

4.2 Results of Linear Regression Analysis

Table 1: model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	0.945	0.894	0.862	13.14721

Source: Author's computation

a. Predictors: (constant)

SHORTLAVERAGEDEBTRATIO, LIQUIDITYRATIO,

CAPITALADEQUACYRATIO

The model explained 89.4% of the variance in performance (R^2 = .894, Adjusted R^2 = .862), which indicates a very strong explanatory power.

ANOVA

model		Sum of Square	Df	Mean Square	F	Sig.
1	Regression	14542.521	3	4847.507	28.045	0.00
	Residual	1728.491	10	172.849		
	Total	16271.012	13			

Source: Author's computation

a. Dependent Variable: PERFORMANCE

b. Predictors: (Constant), SHORTLAVERAGEDEBTRATIO,

LIQUIDITYRATIO, CAPITALADEQUACYRATIO

The ANOVA results show that the regression model is statistically significant ($F(3,10) = 28.045$, $p < .001$), confirming that the predictors jointly explain a significant portion of variation in performance.

Coefficients

		Unstandardized Coefficients		Standardized Coefficients			Collinearity Statistics	
model		B	Beta	Beta	T	Sig.	Tolerance	VIF
1	Constant	27.533	10.277		2.679	0.023		
	Liquidity Ratio	-4.993	3.114	-1.66	-1.604	0.140	0.987	1.013
	Capital Adequacy Ratio	0.012	0.013	0.098	0.932	0.373	0.953	1.049
	Short Leverage Debt Ratio	4.644	0.530	0.929	8.769	0.000	0.946	1.057

Source: Author's computation

a. Dependent Variable: PERFORMANCE

The regression coefficients indicate that liquidity ratio had a negative but non-significant effect on performance (B = –4.993, p = .140). Capital adequacy ratio had a positive but non-significant effect (B = 0.012, p = .373). Short leverage debt ratio had a strong positive and highly significant effect (B = 4.644, p < .001).

The fitted regression model is the giving below:

$FP_t = \beta_0 + \beta_1 LIQR_t + \beta_2 CAR_t + \beta_3 STLDR_t + \varepsilon_t$

$FP_t = 27.533 - (4.993)\ LIQR_t + (0.012)\ CAR_t + (4.644)\ STLDR_t + \varepsilon_t$

Discussion: The regression results reinforce the correlation analysis. Short leverage debt ratio is the only significant predictor of performance, suggesting that firms relying on short-term debt financing experience improved performance. In contrast, liquidity and capital adequacy ratios did not significantly affect performance.

4.3 Correlation Analysis

Pearson correlation was used to assess the relationships among the independent variables (liquidity ratio, capital adequacy ratio, and short leverage debt ratio) and the dependent variable (performance).

Correlations

		Performance	Liquidity Ratio	Capital Adequacy Ratio	Short Laverage Debt Ratio
Performance	Pearson correlation	1	-0.259	0.086	0.925
	Sig. (2-tailed)		0.370	0.769	0.000
	N	14	14	14	14
Liquidity Ratio	Pearson Correlation	-0.259	1	-0.043	-0.096
	Sig. (2-tailed)	0.370		0.884	0.745
	N	14	14	14	14
Capital Adequacy Ratio	Pearson correlation	-0.086	-0.043	1	-0.207
	Sig. (2-tailed)	0.769	0.884		0.478
	N	14	14	14	14
Short Laverage Debt Ratio	Pearson correlation	0.925	-0.096	-0.207	1
	Sig. (2-tailed)	0.000	0.745	0.478	
	N	14	14	14	14

Source: Author's computation

Correlation is significant at the 0.01 level (2-tailed).

The results show that liquidity ratio had a weak, negative, and non-significant relationship with performance ($r = –.259$, $p = .370$). Capital adequacy ratio also had a weak and non-significant negative relationship with performance ($r = –.086$, $p = .769$). Short leverage debt ratio had a very strong positive and significant relationship with performance ($r = .925$, $p < .001$).

Discussion: These results suggest that while liquidity and capital adequacy ratios may not directly enhance performance, reliance on short-term leverage plays a crucial role in improving firm outcomes. This indicates that firms in the sample possibly benefit from short-term debt financing, which may provide resources for growth and profitability.

4.4 Summary of Findings

The analysis shows that while liquidity and capital adequacy ratios have little influence on performance, short leverage debt ratio has a strong and significant positive impact. The overall model demonstrates high explanatory power, highlighting the central role of debt structure in driving firm performance.

CHAPTER FIVE

Summary, Conclusion and Recommendation

5.1 Introduction

This chapter provides a discussion of the study findings in relation to existing literature and theoretical expectations. It also presents the conclusions drawn from the results and offers recommendations for practice, policy, and future research.

5.2 Summary of Findings

The study assessed the impact of credit risk management on the financial performance of AIICO Insurance Plc in Nigeria over the period 2010–2023, using liquidity ratio, capital adequacy ratio, and short leverage debt ratio for credit risk. The analysis employed correlation and multiple regression models, with financial performance measured by Return on Assets (ROA).

5.2.1 Liquidity Ratio and Performance

The study found that liquidity ratio had a negative but insignificant effect on performance. This suggests that higher liquidity did not necessarily improve profitability, and excessive liquidity may reflect idle or underutilized resources. This aligns with theories that caution against over-liquidity (Keynes, 1936; Jensen, 1986) but contrasts with some empirical studies that found a positive influence of liquidity on firm performance (Raheman & Nasr, 2007; Alshatti, 2015).

5.2.2 Capital Adequacy Ratio and Performance

Capital adequacy ratio showed a weak and non-significant relationship with performance. This implies that maintaining high levels of capital adequacy did not directly improve financial performance in AIICO. This finding is consistent with studies that emphasize capital adequacy more as a regulatory and solvency requirement than as a profitability driver (Modigliani & Miller, 1963; Basel Committee on Banking Supervision, 2010).

5.2.3 Short Leverage Debt Ratio and Performance

Short leverage debt ratio exhibited a very strong positive and significant effect on performance. This finding indicates that firms with higher reliance on short-term leverage experienced improved profitability. This supports capital structure theories, such as the trade-off theory, which highlight the benefits of debt in enhancing returns (Myers, 1984; Abor, 2005), but it also raises caution regarding the associated refinancing risks in volatile environments (Demirgüç-Kunt & Maksimovic, 1999).

5.2.4 Overall Model Fit and Robustness

The regression model was statistically significant and explained 89.4% of the variance in performance, indicating that the selected credit risk indicators are strong predictors of AIICO's performance. Importantly, the Variance Inflation Factor (VIF) test confirmed the absence of multicollinearity among the independent variables, ensuring that the regression estimates were stable and reliable (Gujarati & Porter, 2009).

5.3 Conclusion

Based on the findings, the following conclusions are drawn:

1. Liquidity ratio does not significantly affect financial performance, suggesting that excess liquidity may not always be beneficial if not efficiently deployed.
2. Capital adequacy ratio has no significant impact on performance, reinforcing the view that capital buffers serve regulatory and solvency purposes more than profitability enhancement (Basel Committee, 2010).
3. Short leverage debt ratio significantly improves financial performance, making it the most influential credit risk indicator for AIICO Insurance Plc during the study period.
4. The regression model achieved high explanatory power ($R^2 = 0.894$), and the absence of multicollinearity (as confirmed by low VIF values) reinforces the robustness and validity of these results.

Overall, the study concludes that debt structure, particularly short-term leverage, plays a central role in driving financial performance, while liquidity and capital adequacy are less influential for profitability in the Nigerian insurance context.

5.4 Recommendations

5.4.1 For Firms/Management

Firms are encouraged to strategically utilize short-term leverage to enhance performance but should manage associated risks by avoiding excessive debt accumulation.

Excess liquidity should be minimized, with resources reallocated toward productive investments to generate returns.

Capital adequacy should continue to be maintained to satisfy regulatory requirements and ensure solvency, even if its direct effect on profitability is limited.

5.4.2 For Regulators/Policy Makers

NAICOM and other regulators should balance capital adequacy requirements with policies that support firm profitability, especially in the insurance sector.

Regulatory monitoring of short-term leverage practices is essential to ensure that firms benefit from debt without exposing the sector to systemic risk.

5.4.3 For Future Research

Future studies should investigate the long-term sustainability risks associated with short-term leverage dependence.

Expanding the sample size beyond AIICO to include other Nigerian insurers would provide more generalizable insights.

Incorporating addb itional credit risk indicators (e.g., underwriting risk, market risk) could enrich the analysis and offer a broader perspective on firm performance.

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Appendix A: Data Used for Analysis (2010–2023)

The table below presents the financial data extracted from the annual reports of AIICO Insurance Plc which were used to compute the variables for this study.

Year	Total Assets	Total Equity	Current Assets	Current Liabilities	Short-term Borrowings	Net Income	ROA
2010	2.3	280.4	0.18	0.18	0.27	0.18	2.3
2011	1.85	219.4	0.3	0.3	0.39	0.3	1.85
2012	4.84	227.4	0.44	0.44	0.67	0.44	4.84
2013	3.37	206.6	0.62	0.62	0.72	0.62	3.37
2014	1.62	229	0.43	0.43	1.28	0.43	1.62
2015	2.01	187.4	0.018	0.018	0.88	0.018	2.01
2016	1.87	190	0.27	0.27	0.18	0.27	1.87
2017	1.34	201.6	0.39	0.39	0.3	0.39	1.34
2018	1.4	305.4	0.67	0.67	0.44	0.67	1.4
2019	1.68	558.2	0.72	0.72	0.62	0.72	1.68
2020	4.46	674	1.28	1.28	0.43	1.28	4.46
2021	3.23	760.6	0.88	0.88	0.018	0.88	3.23
2022	1.05	891.8	0.51	0.51	0.18	0.51	1.05

2023	1.71	1036.6	0.58	0.58	0.88	4.46	1.28

Source: AIICO Insurance Plc Annual Reports (2010–2023).

Appendix B: Variable Measurement

The variables used in the study were calculated using the following financial ratios:

1. Liquidity Ratio (LIQR)

Liquidity Ratio = Current Assets / Current Liabilities

2. Capital Adequacy Ratio (CAR)

Capital Adequacy Ratio = Total Equity / Total Assets

3. Short Leverage Debt Ratio (STLDR)

Short Leverage Debt Ratio = Short-term Borrowings / Total Assets

4. Financial Performance (ROA)

Return on Assets = Net Income / Total Assets

Appendix C: SPSS Output Summary

Model Summary

R	R Square	Adjusted R Square	Std. Error
0.945	0.894	0.862	13.147

This indicates that 89.4% of the variation in financial performance is explained by the independent variables.

ANOVA Results

Source	Sum of Squares	df	Mean Square	F	Sig

Regression	14542.521	3	4847.507	28.045	0.000
Residual	1728.491	10	172.849		
Total	16271.012	13			

Regression Coefficients

Variable	Coefficient (B)	t-value	Sig
Constant	27.533	2.679	0.023
Liquidity Ratio	-4.993	-1.604	0.140
Capital Adequacy Ratio	0.012	0.932	0.373
Short Leverage Debt Ratio	4.644	8.769	0.000

Appendix D: Regression Model

$FPt = \beta 0 + \beta 1LIQRt + \beta 2CARt + \beta 3STLDRt + \varepsilon t$

Where:

FP = Financial Performance (ROA)

LIQR = Liquidity Ratio

CAR = Capital Adequacy Ratio

STLDR = Short Leverage Debt Ratio

$\beta 0$ = Intercept

$\beta 1 – \beta 3$ = Coefficients of independent variables

ε = Error term